

Just implement it

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 256 megabytes

Given a string S . Just print the frequency of each letter in it.

NOTE: frequency of a letter is how many times it appears in the string.

Input

Only one input line contains a string S .

Output

The number of output lines will be equal to the number of unique letters in the string. In each line, print the letter and its frequency. Note that you have to print the letters in ascending order.

Example

standard input	standard output
abacd	a 2 b 1 c 1 d 1